PROBLEMS

★ **2–1** For a sinusoidal waveform with a peak value of A and a frequency of f_0, use the time average operator to show that the RMS value for this waveform is $A/\sqrt{2}$.

2–2 A function generator produces the periodic voltage waveform shown in Fig. P2–2.

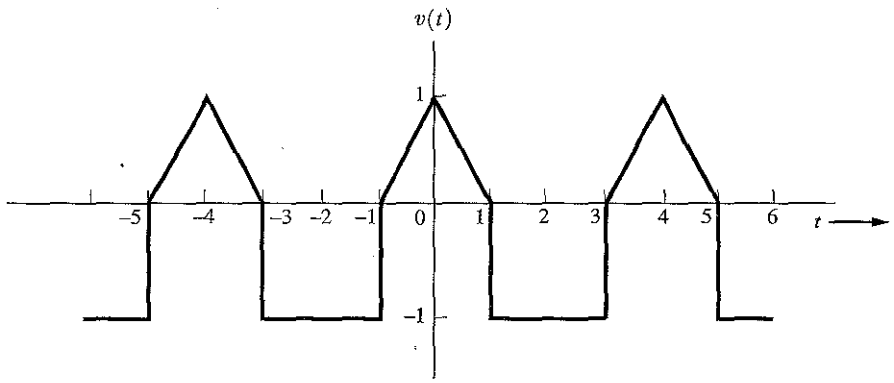

Figure P2–2

(a) Find the value for the DC voltage.
(b) Find the value for the RMS voltage.
(c) If this voltage waveform is applied across a 1,000-Ω load, what is the power dissipated in the load?

2–3 The voltage across a load is given by $v(t) = A_0 \cos \omega_0 t$, and the current through the load is a square wave,

$$i(t) = I_0 \sum_{n=-\infty}^{\infty} \left[\Pi\left(\frac{t - nT_0}{T_0/2}\right) - \Pi\left(\frac{t - nT_0 - (T_0/2)}{T_0/2}\right) \right]$$

where $\omega_0 = 2\pi/T_0$, $T_0 = 1$ sec, $A_0 = 10$ V, and $I_0 = 5$ mA.
(a) Find the expression for the instantaneous power and sketch this result as a function of time.
(b) Find the value of the average power.

★ **2–4** The voltage across a 50-Ω resistive load is the positive portion of a cosine wave. That is,

$$v(t) = \begin{cases} 10 \cos \omega_0 t, & |t - nT_0| < T_0/4 \\ 0, & t \text{ elsewhere} \end{cases}$$

where n is any integer.
(a) Sketch the voltage and current waveforms.
(b) Evaluate the DC values for the voltage and current.
(c) Find the RMS values for the voltage and current.
(d) Find the total average power dissipated in the load.

2–5 For Prob. 2–4, find the energy dissipated in the load during a 1-hr interval if $T_0 = 1$ sec.

2–6 Determine whether each of the following signals is an energy signal or a power signal, and evaluate the normalized energy or power, as appropriate:
(a) $w(t) = \Pi(t/T_0)$.

(b) $w(t) = \Pi(t/T_0) \cos \omega_0 t$.
(c) $w(t) = \cos^2 \omega_0 t$.

2–7 An average-reading power meter is connected to the output circuit of a transmitter. The transmitter output is fed into a 75-Ω resistive load, and the wattmeter reads 67 W.
(a) What is the power in dBm units?
(b) What is the power in dBk units?
(c) What is the value in dBmV units?

2–8 Assume that a waveform with a known RMS value V_{rms} is applied across a 50-Ω load. Derive a formula that can be used to compute the dBm value from V_{rms}.

★ **2–9** An amplifier is connected to a 50-Ω load and driven by a sinusoidal current source, as shown in Fig. P2–9. The output resistance of the amplifier is 10 Ω and the input resistance is 2 kΩ. Evaluate the true decibel gain of this circuit.

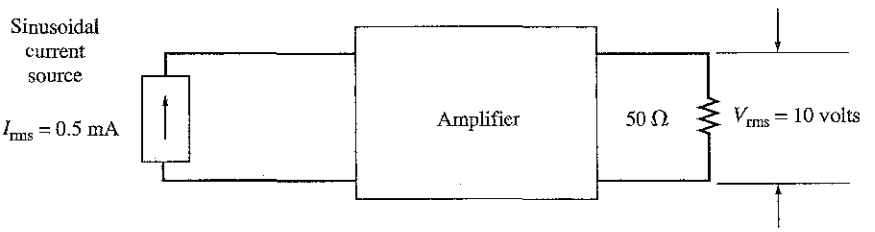

Figure P2–9

★ **2–10** The voltage (RMS) across the 300-Ω antenna input terminals of an FM receiver is 3.5 μV.
(a) Find the input power (watts).
(b) Evaluate the input power as measured in decibels below 1 mW (dBm).
(c) What would be the input voltage (in microvolts) for the same input power if the input resistance were 75 Ω instead of 300 Ω?

2–11 What is the value for the phasor that corresponds to the voltage waveform $v(t) = 12 \sin(\omega_0 t - 25°)$, where $\omega_0 = 2000\pi$?

2–12 A signal is $w(t) = 3 \sin(100\pi t - 30°) + 4 \cos(100\pi t)$. Find the corresponding phasor.

★ **2–13** Evaluate the Fourier transform of

$$w(t) = \begin{cases} e^{-\alpha t}, & t \geq 1 \\ 0, & t < 1 \end{cases}$$

2–14 Find the spectrum for the waveform $w(t) = e^{-\pi(t/T)^2}$ in terms of the parameter T. What can we say about the width of $w(t)$ and $W(f)$ as T increases?

2–15 Using the convolution property, find the spectrum for

$$w(t) = \sin 2\pi f_1 t \cos 2\pi f_2 t$$

★ **2–16** Find the spectrum (Fourier transform) of the triangle waveform

$$s(t) = \begin{cases} At, & 0 < t < T_0 \\ 0, & t \text{ elsewhere} \end{cases}$$

in terms of A and T_0.

2–17 Find the spectrum for the waveform shown in Fig. P2–17.

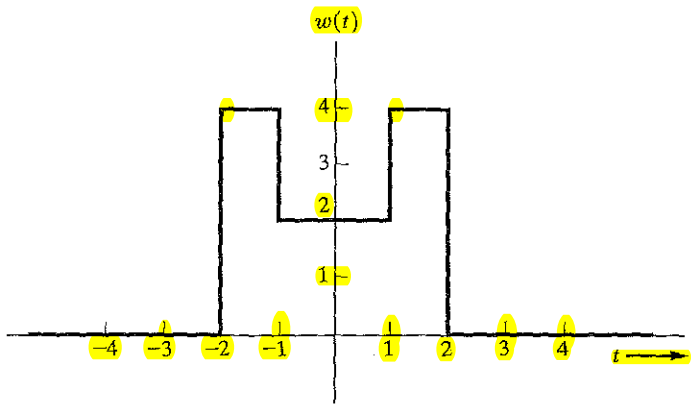

Figure P2–17

2–18 If $w(t)$ has the Fourier transform

$$W(f) = \frac{j2\pi f}{1 + j2\pi f}$$

find $X(f)$ for the following waveforms:

(a) $x(t) = w(2t + 2)$.

(b) $x(t) = e^{-jt}w(t - 1)$.

(c) $x(t) = 2\dfrac{dw(t)}{dt}$.

(d) $x(t) = w(1 - t)$.

2–19 From Eq. (2–30), find $w(t)$ for $W(f) = A\Pi(f/2B)$, and verify your answer by using the duality property.

2–20 Find the quadrature spectral functions $X(f)$ and $Y(f)$ for the damped sinusoidal waveform

$$w(t) = u(t)e^{-at} \sin \omega_0 t$$

where $u(t)$ is a unit step function, $a > 0$, and $W(f) = X(f) + jY(f)$.

2–21 Derive the spectrum of $w(t) = e^{-|t|/T}$.

★ **2–22** Find the Fourier transforms for the following waveforms. Plot the waveforms and their magnitude spectra. [*Hint:* Use Eq. (2–184).]

(a) $\Pi\left(\dfrac{t-3}{4}\right)$.

(b) 2.

(c) $\Lambda\left(\dfrac{t-5}{5}\right)$.

2–23 By using Eq. (2–184), find the approximate Fourier transform for the following waveform:

$$x(t) = \begin{cases} \sin(2\pi t/512) + \sin(70\pi t/512), & 5 < t < 75 \\ 0, & t \text{ elsewhere.} \end{cases}$$

2–24 Evaluate the spectrum for the trapezoidal pulse shown in Fig. P2–24.

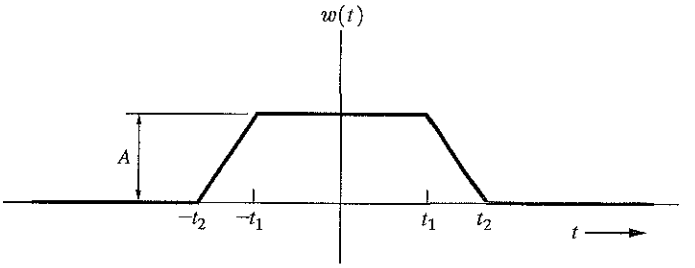

Figure P2–24

2–25 Show that

$$\mathcal{F}^{-1}\{\mathcal{F}[w(t)]\} = w(t)$$

[*Hint:* Use Eq. (2–33).]

2–26 Using the definition of the inverse Fourier transform, show that the value of $w(t)$ at $t=0$ is equal to the area under $W(f)$. That is, show that

$$w(0) = \int_{-\infty}^{\infty} W(f)\, df$$

2–27 Prove that

(a) If $w(t)$ is real and an even function of t, $W(f)$ is real.

(b) If $w(t)$ is real and an odd function of t, $W(f)$ is imaginary.

2–28 Suppose that the spectrum of a waveform as a function of frequency in hertz is

$$W(f) = \frac{1}{2}\,\delta(f-4) + \frac{1}{2}\,\delta(f+4) + \frac{j\pi f}{2 + j2\pi f}\, e^{j\pi f}$$

Find the corresponding spectrum as a function of radian frequency, $W(\omega)$.

2–29 The unit impulse can also be defined as

$$\delta(t) = \lim_{a\to\infty}\left[Ka\left(\frac{\sin at}{at}\right)\right]$$

Find the value of K needed, and show that this definition is consistent with those given in the text. Give another example of an ordinary function such that, in the limit of some parameter, the function becomes a Dirac delta function.

★ **2–30** Use $v(t) = ae^{-at}$, $a > 0$, to approximate $\delta(t)$ as $a \to \infty$.

(a) Plot $v(t)$ for $a = 0.1$, 1, and 10.

(b) Plot $V(f)$ for $a = 0.1$, 1, and 10.

2–31 Show that

$$\operatorname{sgn}(t) \leftrightarrow \frac{1}{j\pi f}$$

[*Hint:* Use Eq. (2–30) and $\int_0^\infty (\sin x)/x\, dx = \pi/2$ from Appendix A.]

2–32 Show that

$$u(t) \leftrightarrow \tfrac{1}{2}\delta(f) + \frac{1}{j2\pi f}$$

[*Hint:* Use the linearity (superposition) theorem and the result of Prob. 2–31.]

Problems

2–33 Show that the sifting property of δ functions, Eq. (2–47), may be generalized to evaluate integrals that involve derivatives of the delta function. That is, show that

$$\int_{-\infty}^{\infty} w(x)\ \delta^{(n)}(x-x_0)dx = (-1)^n w^{(n)}(x_0)$$

where the superscript (n) denotes the nth derivative. (*Hint:* Use integration by parts.)

2–34 Let $x(t) = \Pi\left(\dfrac{t-0.05}{0.1}\right)$. Plot the spectrum of $x(t)$ using MATLAB with the help of Eqs. (2–59) and (2–60). Check your results by using the FFT and Eq. (2–184).

★ **2–35** If $w(t) = w_1(t)w_2(t)$, show that

$$W(f) = \int_{-\infty}^{\infty} W_1(\lambda)W_2(f-\lambda)\ d\lambda$$

where $W(f) = \mathcal{F}[w(t)]$.

2–36 Show that

(a) $\int_{-\infty}^{t} w(\lambda)d\lambda = w(t) * u(t)$.

(b) $\int_{-\infty}^{t} w(\lambda)d\lambda \leftrightarrow (j2\pi f)^{-1}\ W(f) + \frac{1}{2}W(0)\ \delta(f)$.

(c) $w(t) * \delta(t-a) = w(t-a)$.

2–37 Show that

$$\frac{dw(t)}{dt} \leftrightarrow (j2\pi f)W(f)$$

[*Hint:* Use Eq. (2–26) and integrate by parts. Assume that $w(t)$ is absolutely integrable.]

2–38 As discussed in Example 2–8, show that

$$\frac{1}{T}\ \Pi\left(\frac{t}{T}\right) * \Pi\left(\frac{t}{T}\right) = \Lambda\left(\frac{t}{T}\right)$$

2–39 Given the waveform $w(t) = A\Pi(t/T)\ \sin\omega_0 t$, find the spectrum of $w(t)$ by using the multiplication theorem as discussed in Example 2–9.

★ **2–40** Evaluate the following integrals.

(a) $\displaystyle\int_{-\infty}^{\infty} \frac{\sin\ 4\lambda}{4\lambda}\ \delta(t-\lambda)\ d\lambda.$

(b) $\displaystyle\int_{-\infty}^{\infty} (\lambda^3-1)\ \delta(2-\lambda)\ d\lambda.$

2–41 Prove that

$$M(f) * \delta(f-f_0) = M(f-f_0)$$

2–42 Evaluate $y(t) = w_1(t) * w_2(t)$, where

$$w_1(t) = \begin{cases} 1, & |t| < T_0 \\ 0, & t \text{ elsewhere} \end{cases}$$

and

$$w_2(t) = \begin{cases} [1-2|t|], & |t| < \frac{1}{2}T_0 \\ 0, & t \text{ elsewhere} \end{cases}$$

2–43 Given $w(t) = 5 + 12 \cos \omega_0 t$, where $f_0 = 10$ Hz, find
(a) $R_w(\tau)$.
(b) $\mathcal{P}_w(f)$.

2–44 Given the waveform

$$w(t) = A_1 \cos(\omega_1 t + \theta_1) + A_2 \cos(\omega_2 t + \theta_2)$$

where A_1, A_2, ω_1, ω_2, θ_1, and θ_2 are constants,
(a) Find the autocorrelation for $w(t)$ as a function of the constants.
(b) Find the PSD function for $w(t)$.
(c) Sketch the PSD for the case of $\omega_1 \neq \omega_2$.
(d) Sketch the PSD for the case of $\omega_1 = \omega_2$ and $\theta_1 = \theta_2 + 90°$.
(e) Sketch the PSD for the case of $\omega_1 = \omega_2$ and $\theta_1 = \theta_2$.

★ **2–45** Given the periodic voltage waveform shown in Fig. P2–45,
(a) Find the DC value for this waveform.
(b) Find the RMS value for this waveform.
(c) Find the complex exponential Fourier series.
(d) Find the voltage spectrum for this waveform.

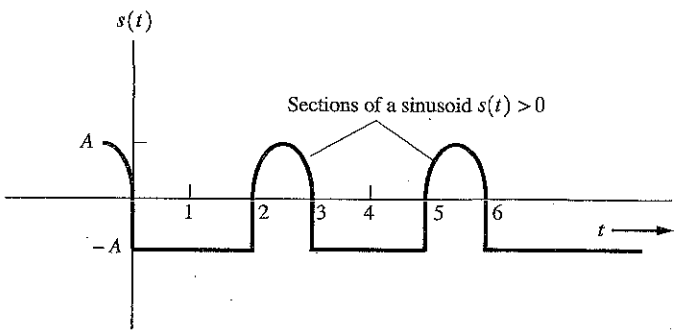

Figure P2–45

2–46 Determine if $s_1(t)$ and $s_2(t)$ are orthogonal over the interval $(-\frac{5}{2}T_2 < t < \frac{5}{2}T_2)$, where $s_1(t) = A_1 \cos(\omega_1 t + \varphi_1)$, $s_2(t) = A_2 \cos(\omega_2 t + \varphi_2)$, and $\omega_2 = 2\pi/T_2$ for the following cases.
(a) $\omega_1 = \omega_2$ and $\varphi_1 = \varphi_2$.
(b) $\omega_1 = \omega_2$ and $\varphi_1 = \varphi_2 + \pi/2$.
(c) $\omega_1 = \omega_2$ and $\varphi_1 = \varphi_2 + \pi$.
(d) $\omega_1 = 2\omega_2$ and $\varphi_1 = \varphi_2$.
(e) $\omega_1 = \frac{4}{5}\omega_2$ and $\varphi_1 = \varphi_2$.
(f) $\omega_1 = \pi\omega_2$ and $\varphi_1 = \varphi_2$.

★ **2–47** Let $s(t) = A_1 \cos(\omega_1 t + \varphi_1) + A_2 \cos(\omega_2 t + \varphi_2)$. Determine the RMS value of $s(t)$ in terms of A_1 and A_2 for the following cases.
(a) $\omega_1 = \omega_2$ and $\varphi_1 = \varphi_2$.
(b) $\omega_1 = \omega_2$ and $\varphi_1 = \varphi_2 + \pi/2$.
(c) $\omega_1 = \omega_2$ and $\varphi_1 = \varphi_2 + \pi$.
(d) $\omega_1 = 2\omega_2$ and φ_1 and φ_2.
(e) $\omega_1 = 2\omega_2$ and $\varphi_1 = \varphi_2 + \pi$.

2–48 Show that

$$\sum_{k=-\infty}^{\infty} \delta(t-kT_0) \leftrightarrow f_0 \sum_{k=-\infty}^{\infty} \delta(f-nf_0)$$

where $f_0 = 1/T_0$. [*Hint:* Expand $\sum_{k=-\infty}^{\infty} \delta(t-kT_0)$ into a Fourier series and then take the Fourier transform.]

★ **2–49** Three functions are shown in Fig. P2–49.

(a) Show that these functions are orthogonal over the interval (−4, 4).

(b) Find the corresponding orthonormal set of functions.

(c) Expand the waveform

$$w(t) = \begin{cases} 1, & 0 \le t \le 4 \\ 0, & t \text{ elsewhere} \end{cases}$$

into an orthonormal series by using the orthonormal set found in part (b).

(d) Evaluate the mean square error for the orthogonal series obtained in part (c) by evaluating

$$\varepsilon = \int_{-4}^{4} \left[w(t) - \sum_{j=1}^{3} a_j \varphi_j(t) \right]^2 dt$$

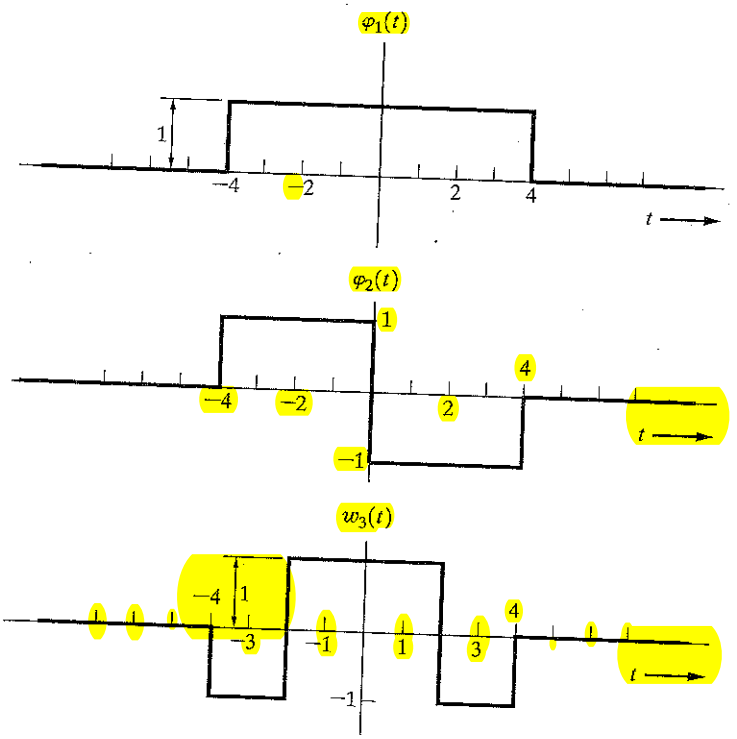

Figure P2–49

(e) Repeat parts (c) and (d) for the waveform

$$w(t) = \begin{cases} \cos(\frac{1}{4}\pi t), & -4 \le t \le 4 \\ 0, & t \text{ elsewhere} \end{cases}$$

Are the three orthonormal functions a complete orthonormal set?

2–50 Show that the quadrature Fourier series basis functions $\cos(n\omega_0 t)$ and $\sin(n\omega_0 t)$, as given in Eq. (2–95), are orthogonal over the interval $a < t < a + T_0$, where $\omega_0 = 2\pi/T_0$.

2–51 Find expressions for the complex Fourier series coefficients that represent the waveform shown in Fig. P2–51.

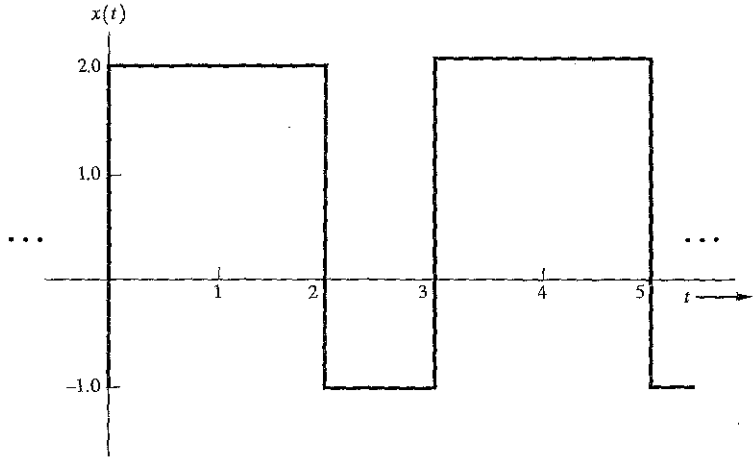

Figure P2–51

2–52 The periodic signal shown in Fig. P2–51 is passed through a linear filter having the impulse response $h(t) = e^{-\alpha t}u(t)$, where $t > 0$ and $\alpha > 0$.

(a) Find expressions for the complex Fourier series coefficients associated with the output waveform $y(t) = x(t) * h(t)$.

(b) Find an expression for the normalized power of the output, $y(t)$.

2–53 Find the complex Fourier series for the periodic waveform given in Figure P2–2.

★ **2–54** Find the complex Fourier series coefficients for the periodic rectangular waveform shown in Fig. P2–54 as a function of A, T, b, and τ_0. [*Hint:* The answer can be reduced to a $(\sin x)/x$ form multiplied by a phase-shift factor, $e^{j\theta_n(\tau_0)}$.]

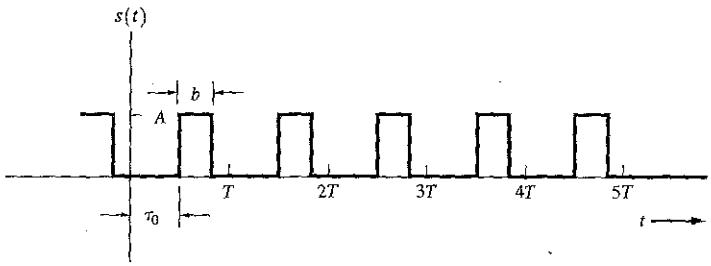

Figure P2–54

2–55 For the waveform shown in Fig. P2–55, find
(a) The complex Fourier series.
(b) The quadrature Fourier series.

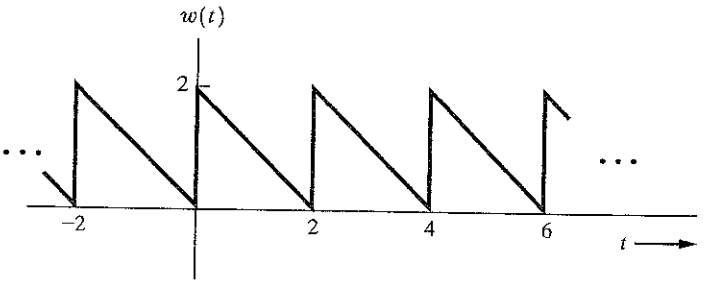

Figure P2-55

★ **2–56** Given a periodic waveform $s(t) = \sum_{n=-\infty}^{\infty} p(t - nT_0)$, where

$$p(t) = \begin{cases} At, & 0 < t < T \\ 0, & t \text{ elsewhere} \end{cases}$$

and $T \leq T_0$,
(a) Find the c_n Fourier series coefficients.
(b) Find the $\{x_n, y_n\}$ Fourier series coefficients.
(c) Find the $\{D_n, \varphi_n\}$ Fourier series coefficients.

2–57 Prove that the polar form of the Fourier series, Eq. (2–103), can be obtained by rearranging the terms in the complex Fourier series, Eq. (2–88).

2–58 Prove that Eq. (2–93) is correct.

2–59 Let two complex numbers c_1 and c_2 be represented by $c_1 = x_1 + jy_1$ and $c_2 = x_2 + jy_2$, where x_1, x_2, y_1 and y_2 are real numbers. Show that $\mathrm{Re}\{\cdot\}$ is a linear operator by demonstrating that

$$\mathrm{Re}\{c_1 + c_2\} = \mathrm{Re}\{c_1\} + \mathrm{Re}\{c_2\}$$

2–60 Assume that $y(t) = s_1(t) + 2s_2(t)$, where $s_1(t)$ is given by Fig. P2–45 and $s_2(t)$ is given by Fig. P2–54. Let $T = 3$, $b = 1.5$, and $\tau_0 = 0$. Find the complex Fourier coefficients $\{c_n\}$ for $y(t)$.

2–61 Evaluate the PSD for the waveform shown in Fig. P2–2.

★ **2–62** Assume that $v(t)$ is a triangular waveform, as shown in Fig. P2–62.

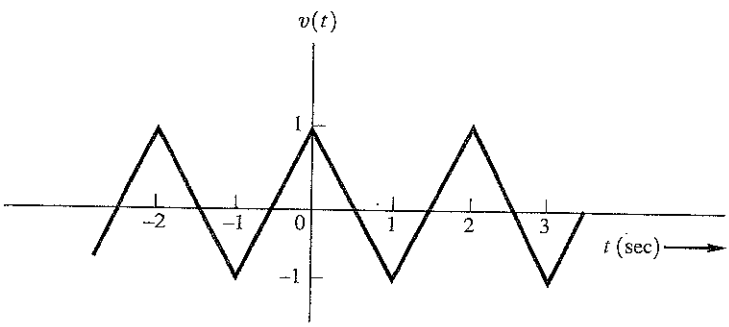

Figure P2–62

(a) Find the complex Fourier series for $v(t)$.
(b) Calculate the normalized average power.
(c) Calculate and plot the voltage spectrum.
(d) Calculate and plot the PSD.

2–63 Let any complex number be represented by $c = x + jy$, where x and y are real numbers. If c^* denotes the complex conjugate of c, show that

(a) $\text{Re}\{c\} = \frac{1}{2}c + \frac{1}{2}c^*$.

(b) $\text{Im}\{c\} = \dfrac{1}{2j}c - \dfrac{1}{2j}c^*$.

Note that when $c = e^{jz}$, part (a) gives the definition of $\cos z$ and part (b) gives the definition of $\sin z$.

★ **2–64** Calculate and plot the PSD for the half-wave rectified sinusoid described in Prob. 2–4.

2–65 The basic definitions for sine and cosine waves are

$$\sin z_1 \triangleq \frac{e^{jz_1} - e^{-jz_1}}{2j} \quad \text{and} \quad \cos z_2 \triangleq \frac{e^{jz_2} + e^{-jz_2}}{2}.$$

Show that

(a) $\cos z_1 \cos z_2 = \frac{1}{2}\cos(z_1 - z_2) + \frac{1}{2}\cos(z_1 + z_2)$.

(b) $\sin z_1 \sin z_2 = \frac{1}{2}\cos(z_1 - z_2) - \frac{1}{2}\cos(z_1 + z_2)$.

(c) $\sin z_1 \cos z_2 = \frac{1}{2}\sin(z_1 - z_2) + \frac{1}{2}\sin(z_1 + z_2)$.

2–66 Let two complex numbers be respectively represented by $c_1 = x_1 + jy_1$ and $c_2 = x_2 + jy_2$, where x_1, x_2, y_1, and y_2 are real numbers. Show that

$$\text{Re}\{c_1\}\,\text{Re}\{c_2\} = \tfrac{1}{2}\,\text{Re}\{c_1 c_2^*\} + \tfrac{1}{2}\,\text{Re}\{c_1 c_2\}$$

Note that this is a generalization of the $\cos z_1 \cos z_2$ identity of Prob. 2–65, where, for the $\cos z_1 \cos z_2$ identity, $c_1 = e^{jz_1}$ and $c_2 = e^{jz_2}$.

2–67 Prove that the Fourier transform is a linear operator. That is, show that

$$\mathcal{F}[ax(t) + by(t)] = a\mathcal{F}[x(t)] + b\mathcal{F}[y(t)]$$

2–68 Plot the amplitude and phase response for the transfer function

$$H(f) = \frac{j10f}{5 + jf}$$

2–69 Given the filter shown in Fig. P2–69, where $w_1(t)$ and $w_2(t)$ are voltage waveforms,

(a) Find the transfer function.
(b) Plot the magnitude and phase response.
(c) Find the power transfer function.
(d) Plot the power transfer function.

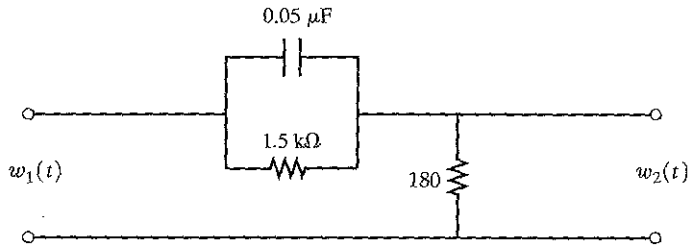

Figure P2–69

2–70 A signal with a PSD of

$$\mathcal{P}_x(f) = \frac{2}{(1/4\pi)^2 + f^2}$$

is applied to the network shown in Fig. P2–70.
(a) Find the PSD for $y(t)$.
(b) Find the average normalized power for $y(t)$.

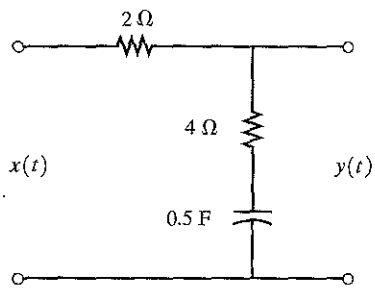

Figure P2–70

2–71 A signal $x(t)$ has a PSD

$$\mathcal{P}_x(f) = \frac{K}{[1 + (2\pi f/B)^2]^2}$$

where $K > 0$ and $B > 0$.
(a) Find the 3-dB bandwidth in terms of B.
(b) Find the equivalent noise bandwidth in terms of B.

★ **2–72** The signal $x(t) = e^{-400\pi t}u(t)$ is applied to a brick-wall low-pass filter whose transfer function is

$$H(f) = \begin{cases} 1, & |f| \le B \\ 0, & |f| > B \end{cases}$$

Find the value of B such that the filter passes one-half the energy of $x(t)$.

2–73 Show that the average normalized power of a waveform can be found by evaluating the autocorrelation $R_w(\tau)$ at $\tau = 0$. That is, $P = R_w(0)$.
[*Hint:* See Eqs. (2–69) and (2–70).]

2–74 The signal $x(t) = 0.5 + 1.5\cos[(\frac{2}{3})\pi t] + 0.5\sin[(\frac{2}{3})\pi t]$ V is passed through an RC low-pass filter (see Fig. 2–15a) where $R = 1\ \Omega$ and $C = 1$ F.
(a) What is the input PSD, $\mathcal{P}_x(f)$?
(b) What is the output PSD, $\mathcal{P}_y(f)$?
(c) What is the average normalized output power, P_y?

2–75 The input to the RC low-pass filter shown in Fig. 2–15 is

$$x(t) = 0.5 + 1.5\cos\omega_x t + 0.5\sin\omega_x t$$

Assume that the cutoff frequency is $f_0 = 1.5f_x$.
(a) Find the input PSD, $\mathcal{P}_x(f)$.
(b) Find the output PSD, $\mathcal{P}_y(f)$
(c) Find the normalized average power of the output, $y(t)$.

★ **2–76** Using MATLAB, plot the frequency magnitude response and phase response for the low-pass filter shown in Fig. P2–76, where $R_1 = 7.5$ kΩ, $R_2 = 15$ kΩ, and $C = 0.1$ μF.

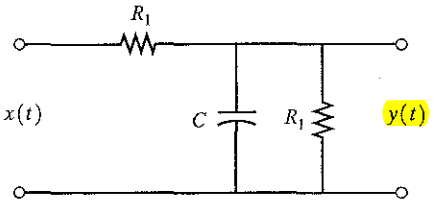

Figure P2–76

2–77 A comb filter is shown in Fig. P2–77. Let $T_d = 0.1$.
(a) Plot the magnitude of the transfer function for this filter.
(b) If the input is $x(t) = \Pi(t/T)$, where $T = 1$, plot the spectrum of the output $|Y(f)|$.

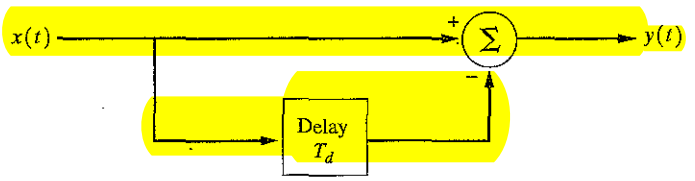

Figure P2–77

2–78 A signal $x(t) = \Pi(t - 0.5)$ passes through a filter that has the transfer function $H(f) = \Pi(f/B)$. Plot the output waveform when
(a) $B = 0.6$ Hz.
(b) $B = 1$ Hz.
(c) $B = 50$ Hz.

★ **2–79** Examine the distortion effect of an *RC* low-pass filter. Assume that a unity-amplitude periodic square wave with a 50% duty cycle is present at the filter input and that the filter has a 3-dB bandwidth of 1,500 Hz. Using a computer, find and plot the output waveshape if the square wave has a frequency of
(a) 300 Hz.
(b) 500 Hz.
(c) 1,000 Hz.
(*Hint:* Represent the square wave with a Fourier series.)

2–80 Given that the PSD of a signal is flat [i.e., $\mathcal{P}_s(f) = 1$], design an *RC* low-pass filter that will attenuate this signal by 20 dB at 15 kHz. That is, find the value for the *RC* of Fig. 2–15 such that the design specifications are satisfied.

2–81 The bandwidth of $g(t) = e^{-0.1t}$ is approximately 0.5 Hz; thus, the signal can be sampled with a sampling frequency of $f_s = 1$ Hz without significant aliasing. Take samples a_n over the time interval (0, 14). Use the sampling theorem, Eq. (2–158), to reconstruct the signal. Plot and compare the reconstructed signal with the original signal. Do they match? What happens when the sampling frequency is reduced?

★ **2–82** A waveform, $20 + 20\sin(500t + 30°)$, is to be sampled periodically and reproduced from these sample values.
(a) Find the maximum allowable time interval between sample values.
(b) How many sample values need to be stored in order to reproduce 1 sec of this waveform?

2–83 Using a computer program, calculate the DFT of a rectangular pulse, $\Pi(t)$. Take five samples of the pulse and pad it with 59 zeros so that a 64-point FFT algorithm can be used. Sketch the resulting magnitude spectrum. Compare this result with the actual spectrum for the pulse. Try other combinations of the number of pulse samples and zero-pads to see how the resulting FFT changes.

★ **2–84** Using the DFT, compute and plot the spectrum of $\Lambda(t)$. Check your results against those given in Fig. 2–6c.

2–85 Using the DFT, compute and plot $|W(f)|$ for the pulse shown in Fig. P2–24, where $A = 1$, $t_1 = 1$s, and $t_2 = 2$s.

2–86 Let a certain waveform be given by

$$w(t) = 4\sin(2\pi f_1 t + 30°) + 2\cos(2\pi f_2 t - 10°),$$

where $f_1 = 10$ Hz and $f_2 = 25$ Hz,

(a) Using the DFT, compute and plot $|W(f)|$ and $\theta(f)$

(b) Let $\mathcal{P}_w(f)$ denote the PSD of $w(t)$. Using the DFT, compute and plot $\mathcal{P}_w(f)$.

(c) Check your computed results obtained in parts (a) and (b) against known correct results that you have evaluated by analytical methods. 0

2–87 Using the DFT, compute and plot $|S(f)|$ for the periodic signal shown in Fig. P2–45, where $A = 5$.

2–88 The transfer function of a raised cosine-rolloff filter is

$$H(f) = \begin{cases} 0.5[1 + \cos(0.5\pi f/f_0)], & |f| \le 2f_0 \\ 0, & f \text{ elsewhere} \end{cases}$$

Let $f_0 = 1$ Hz. Using the IFFT, compute the impulse response $h(t)$ for this filter. Compare your computed results with those shown in Fig. 3–26b for the case of $r = 1$.

2–89 Given the low-pass filter shown in Fig. 2–15,

(a) Find the equivalent bandwidth in terms of R and C.

(b) Find the first zero-crossing (null) bandwidth of the filter.

(c) Find the absolute bandwidth of the filter.

2–90 Assume that the PSD of a signal is given by

$$\mathcal{P}_s(f) = \left[\frac{\sin(\pi f/B_n)}{\pi f/B_n}\right]^2$$

where B_n is the null bandwidth. Find the expression for the equivalent bandwidth in terms of the null bandwidth.

2–91 Table 2–5 gives the bandwidths of a BPSK signal in terms of six different definitions. Using these definitions and Eq. (2–201), show that the results given in the table are correct.

★ **2–92** Given a triangular pulse signal

$$s(t) = \Lambda(t/T_0)$$

(a) Find the absolute bandwidth of the signal.

(b) Find the 3-dB bandwidth in terms of T_0.

(c) Find the equivalent bandwidth in terms of T_0.

(d) Find the zero-crossing bandwidth in terms of T_0.